

CALDER AGENTIC AI ENGINEERING WEEK 3 REPORT

1. WEEK 3 CONCEPTS

Embedding: An embedding is a numerical vector representation of text that captures its meaning. Similar concepts have similar vectors placed close together in vector space.

Cosine Similarity: Cosine similarity measures how similar two vectors are by calculating the angle between them.

Values:

1.0: Very similar (same direction)

0.0: Not similar (perpendicular)

-1.0: Opposite (opposite direction)

Dimensionality: Dimensionality refers to the number of values in a vector representation.

Why Dimensionality Matters:

- More dimensions = More information captured
- More dimensions = More storage required
- More dimensions = Slower computation

Vector Database: A specialized database designed to store and search vectors efficiently. It finds the most similar vectors to your query.

i- ChromaDB

Features:

- In-memory or persistent storage
- Metadata filtering
- Easy integration with LangChain

ii- FAISS (Facebook AI Similarity Search)

Features:

- In-process (no server required)
- Very fast
- GPU acceleration support

RAG: RAG stands for Retrieval-Augmented Generation. It retrieves relevant documents and uses them as context for the LLM to generate better answers.

Naive RAG Pipeline

INGESTION PHASE:

1. Load Documents
2. Split into Chunks
3. Create Embeddings
4. Store in Vector Database

QUERY PHASE:

5. User Asks Question
6. Embed Question
7. Retrieve Relevant Chunks
8. Send to LLM with Context
9. Generate Response

Text Splitting (Chunking): Dividing documents into smaller pieces called chunks.

Advanced Retrieval:

i- Hybrid Search

Combining keyword-based search (BM25) with semantic search.

Why Hybrid:

- Keywords catch exact matches
- Semantics catch meaning-based matches
- Better overall coverage

ii- BM25 (Best Matching 25)

A ranking function for keyword search based on term frequency and inverse document frequency.

iii- **EnsembleRetriever**

Combines multiple retrievers and returns results from both.

iv- **Cross-Encoder Re-ranking**

A model that takes a query and a document together and scores their relevance directly.

v- **HyDE (Hypothetical Document Embeddings)**

Generates a hypothetical document that would answer the question, then uses that document's embedding for retrieval.

vi- **Multi-Query Retrieval**

Generates multiple variations of the user's question and retrieves for each.

RAG Evaluation:

RAGAS Metrics

RAGAS is a framework for evaluating RAG systems without human labels.

1. Faithfulness

Whether the generated answer is factually supported by the retrieved context.

2. Answer Relevancy

Whether the answer directly addresses the question.

3. Context Precision

Whether the retrieved chunks are actually relevant to the question.

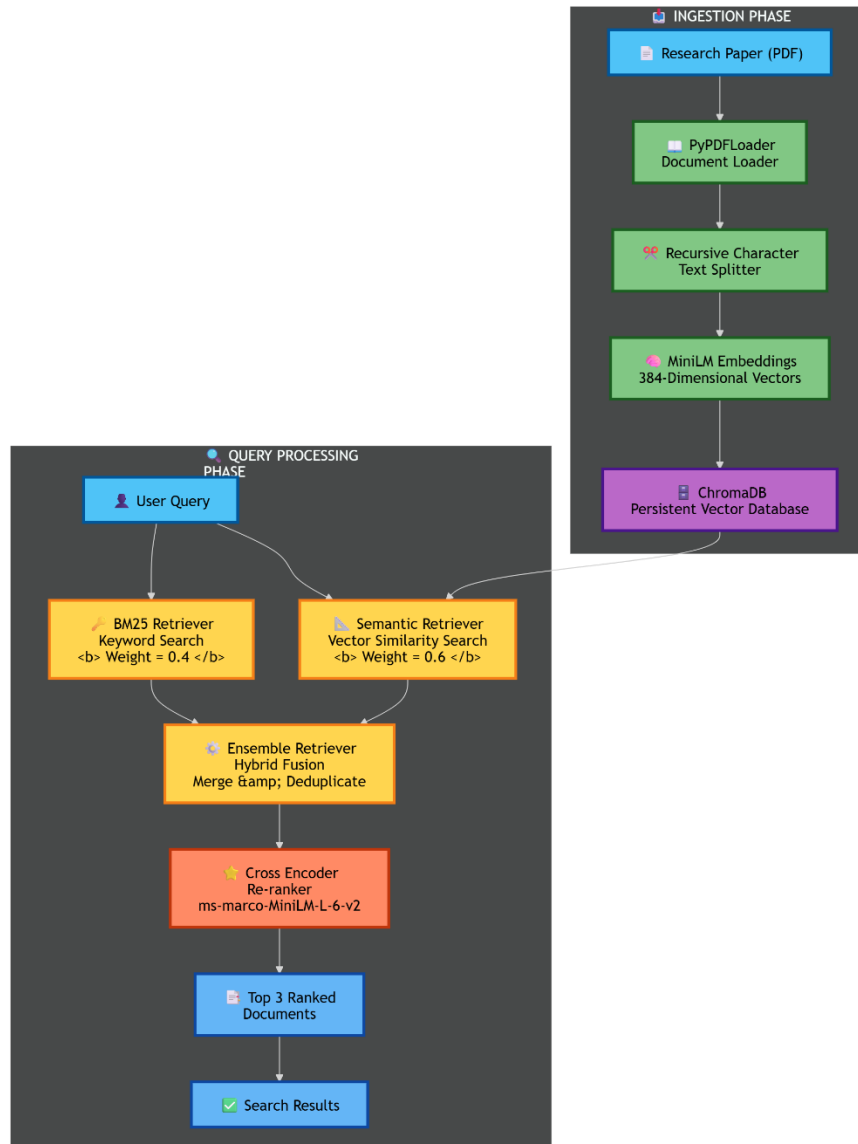
4. Context Recall

Whether the retrieved chunks contain all information needed to answer.

2. INTERMEDIATE PROJECT

I have chosen option C as my intermediate project (Hybrid search engine).

i. System architecture



ii. Technology stack

Component	Technology	Purpose
Document Loader	PyPDFLoader	Load PDF documents
Text Splitter	RecursiveCharacterTextSplitter	Split into 512-token chunks with 50-token overlap
Embedding Model	all-MiniLM-L6-v2	Convert text to 384-dim vectors
Vector Database	ChromaDB	Store and query embeddings
Keyword Search	BM25Retriever	Exact keyword matching
Hybrid Retriever	EnsembleRetriever	Combine BM25 + Semantic
Re-ranker	Cross-Encoder (ms-marco-MiniLM-L-6-v2)	Deep relevance scoring
Evaluation	Custom metrics	Context Precision, Answer Relevancy

iii. Week 3 concepts applied:

Concept: Sentence-Transformer Embeddings

Every document chunk is converted into a 384-dimensional embedding vector using the all-MiniLM-L6-v2 model.

Concept: ChromaDB Storage + Document Chunking

Documents are split into 512-character chunks with 50-character overlap, then stored in ChromaDB with metadata.

Concept: Full RAG Pipeline

Pipeline Steps:

Load → Split → Embed → Store → Retrieve → Generate

Concept: Hybrid Search (BM25 + Semantic)

Combines BM25 keyword search with semantic retrieval using EnsembleRetriever.

Concept: Cross-Encoder Re-ranking

After initial retrieval, a cross-encoder (ms-marco-MiniLM-L-6-v2) re-ranks the results for deeper relevance scoring.

Concept: Evaluation Metrics

Measures whether retrieved chunks are relevant to the query.

- iv. Error handling and fault tolerance

1. Missing or Corrupted PDF (Step 1 & 2):

Scenario: PDF fails to download or is corrupted

Action: Pipeline catches HTTP/IO error and immediately aborts

2. Chunking Failure (Step 3):

Scenario: PDF loaded but 0 chunks produced (e.g., image-only PDF)

Action: Pipeline detects 0 chunks and raises PipelineError

3. Embedding Model Failure (Step 4):

Scenario: HuggingFace Hub is down or offline

Action: Exception is caught, pipeline halts safely

4. Vector Store Failure (Step 5):

Scenario: Read/Write permission error on disk

Action: Caught and aborted safely

- v. Features & Screenshots

Feature 1: PDF Download & Processing

```
(calderr-env) PS C:\Users\USER\Desktop\calderr-ai-2026> python "WEEK 3\intermediate_project\hybrid_search_engine.py"
>> \
C:\Users\USER\Desktop\calderr-ai-2026\WEEK 3\intermediate_project\hybrid_search_engine.py:27: DeprecationWarning: `langchain-community` is being sunset and is no longer actively maintained. See https://github.com/langchain-ai/langchain-community/issues/674 for details and migration guidance toward standalone integration packages.
  from langchain_community.retrievers import BM25Retriever
=====
PRODUCTION-GRADE HYBRID SEARCH ENGINE
=====

[STEP 1] Downloading 'attention is all you need' PDF...
  PDF already exists: C:\Users\USER\Desktop\calderr-ai-2026\WEEK 3\intermediate_project\docs\attention_is_all_you_need.pdf

[STEP 2] Loading Document from C:\Users\USER\Desktop\calderr-ai-2026\WEEK 3\intermediate_project\docs\attention_is_all_you_need.pdf...
  Successfully loaded 15 pages.

[STEP 3] Splitting Documents (chunk_size=512, overlap=50)...
  Created 92 chunks.
```

Feature 2: Hybrid Retriever Construction

```
[STEP 4] Loading Embedding Model (all-MiniLM-L6-v2)...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
  Embedding model loaded successfully!

[STEP 5] Storing in ChromaDB (C:\Users\USER\Desktop\calderr-ai-2026\WEEK 3\intermediate_project\chroma_hybrid_db)...
  Stored 92 chunks in ChromaDB

[STEP 6] Building Hybrid Retriever (BM25=0.4, Semantic=0.6)...
  Hybrid retriever ready!

[STEP 7] Loading Cross-Encoder Re-ranker (cross-encoder/ms-marco-MiniLM-L-6-v2)...
Loading weights: 100%|████████████████████████████████████████| 105/105 [00:00<00:00, 4995.88it/s]
  Re-ranker loaded successfully!
```

Feature 3: Evaluation on "Attention Is All You Need" Paper

```

=====
RUNNING EVALUATION ON 'ATTENTION IS ALL YOU NEED' PAPER
=====
Q01 [PASS] P:0.00 R:1.00 | What is the primary mechanism replacing recurrence in this architecture?
Q02 [PASS] P:0.00 R:1.00 | What is the name of the new network architecture introduced?
Q03 [PASS] P:0.33 R:1.00 | What is the formula for scaled dot-product attention?
Q04 [PASS] P:0.00 R:1.00 | How many identical layers does the encoder stack have?
Q05 [FAIL] P:0.00 R:0.00 | What optimization algorithm was used during training?
Q06 [FAIL] P:0.33 R:0.00 | What prevents leftward information flow in the decoder?
Q07 [PASS] P:0.00 R:1.00 | What is used to inject sequence order information since there are no recurrences?
Q08 [PASS] P:0.33 R:1.00 | How many attention heads (h) were used in the base model?
Q09 [FAIL] P:0.33 R:0.33 | What dataset was used for English-to-German translation?
Q10 [FAIL] P:0.00 R:0.00 | What regularization technique is applied to the output of each sub-layer?

-----
Avg Context Precision: 0.13
Avg Answer Relevancy: 0.63
=====

```

Feature 4: Interactive Search

```

Query: What is the primary mechanism replacing recurrence in this architecture

Found 11 results. Top 3 after re-ranking:

[1] (Score: 3.26)
    tion models in various tasks, allowing modeling of dependencies without regard to their distance in
the input or output sequences [2, 19]. In all but a few cases [27], however, such attention mechanis...

[2] (Score: 1.06)
    End-to-end memory networks are based on a recurrent attention mechanism instead of sequence-
aligned recurrence and have been shown to perform well on simple-language question answering and
language m...

[3] (Score: 0.85)
    7 Conclusion
In this work, we presented the Transformer, the first sequence transduction model based entirely on
attention, replacing the recurrent layers most commonly used in encoder-decoder archite...

Query: █

```

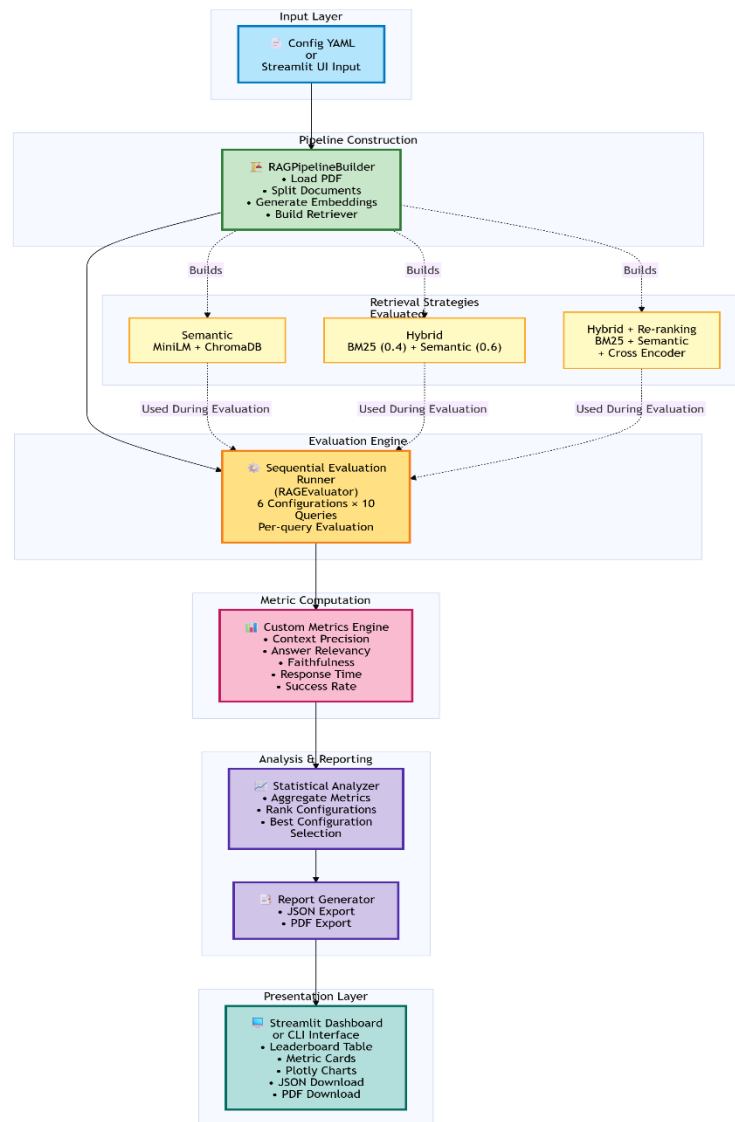
3. PRODUCTION PROJECT

I have chosen option 3-p-c for my production project (**RAG Evaluation Benchmark**).

Option C include:

- Systematic testing across 6 configurations
- Statistical comparison of results
- Publishable report generation
- Reusable framework for future projects

i- System architecture



ii- Technology Stack

Component	Technology	Purpose
Document Loader	PyPDFLoader	Load PDF documents
Text Splitter	RecursiveCharacterTextSplitter	Split into configurable chunks
Embedding Model	all-MiniLM-L6-v2	Convert text to 384-dim vectors
Vector Database	ChromaDB	Store embeddings per config
Keyword Search	BM25Retriever	Exact keyword matching (weight: 0.4)
Semantic Search	ChromaDB Retriever	Meaning-based matching (weight: 0.6)
Hybrid Retriever	EnsembleRetriever	Combine BM25 + Semantic
Re-ranker	Cross-Encoder (ms-marco-MiniLM-L-6-v2)	Deep relevance scoring
UI Framework	Streamlit	Interactive dashboard
Visualization	Plotly	Charts and graphs
Data Processing	Pandas	Data manipulation
Evaluation	Custom Metrics	Context Precision, Answer Relevancy, Faithfulness

iii- Week 3 concepts applied

Concept: Sentence-Transformer Embeddings (all-MiniLM-L6-v2)

Every document chunk is converted into a 384-dimensional embedding vector for semantic retrieval.

Concept: ChromaDB Storage & Document Chunking

Tested with multiple chunk sizes (512/50, 1024/100) and stored in ChromaDB with collection-per-config isolation.

Concept: Full RAG Pipeline

Pipeline Steps:

Load → Split → Embed → Store → Retrieve → Evaluate

Concept: Hybrid Search (BM25 + Semantic)

Combines BM25 keyword search with semantic retrieval using EnsembleRetriever.

Concept: Cross-Encoder Re-ranking

After initial retrieval, a cross-encoder re-ranks results for deeper relevance scoring.

Concept: RAGAS Metrics (Custom Implementation)

1. Context Precision

Measures whether retrieved chunks are relevant to the query.

2. Answer Relevancy

Measures whether retrieved content actually answers the question.

3. Faithfulness

Measures whether the answer is factually supported by the context.

iv- Error handling & Fault tolerance

1. Missing or Corrupted PDF:

Scenario: PDF fails to download or is corrupted

Action: Pipeline catches HTTP/IO error and immediately aborts

2. Chunking Failure:

Scenario: PDF loaded but 0 chunks produced (image-only PDF)

Action: Pipeline detects 0 chunks and raises PipelineError

3. Embedding Model Failure:

Scenario: HuggingFace Hub is down or model name invalid

Action: Exception caught, pipeline halts safely

4. Vector Store Failure:

Scenario: Read/Write permission error on disk

Action: Caught and aborted safely

5. Unknown Retrieval Strategy:

Scenario: Config has unrecognized retrieval_strategy value

Action: PipelineError raised with descriptive message

6. Per-Query Failures (Graceful Degradation):

Scenario: Single evaluation query fails (LLM timeout)

Action: Error logged, query skipped, evaluation continues

V- Features & Screenshots

Feature 1: CLI Benchmark Mode

```
PS C:\Users\USER\Desktop\calder-ai-2026> .\calder-ai-Scripts\Activate.ps1  
(calder-emv) PS C:\Users\USER\Desktop\calder-ai-2026> python "WEEK 3/production_project/project3_p_c_rag_benchmark.py"  
C:\Users\USER\Desktop\calder-ai-2026\WEEK 3\production_project\project3_p_c_rag_benchmark.py:37: DeprecationWarning: `langchain-community` is being sunset and is no longer actively maintained. See https://  
github.com/langchain-ai/langchain-community/issues/674 for details and migration guidance toward standalone integration packages.  
    from langchain_community.retrievers import BM25Retriever  
  
-----  
RAG EVALUATION BENCHMARK - PRODUCTION MODE  
-----  
  
[1/6] Evaluating Configuration: Semantic_512_3  
  
Evaluating: Semantic_512_3  
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HuggingFace_TOKEN to enable higher rate limits and faster downloads.  
Loading weights: 100%|███████████████████████████████████████| 103/103 [00:00<00:00, 2152.581t/s]  
  
[2/6] Evaluating Configuration: Semantic_512_5  
  
Evaluating: Semantic_512_5  
Loading weights: 100%|███████████████████████████████████████| 103/103 [00:00<00:00, 5758.411t/s]  
  
[3/6] Evaluating Configuration: Hybrid_512_3  
  
Evaluating: Hybrid_512_3  
Loading weights: 100%|███████████████████████████████████████| 103/103 [00:00<00:00, 6060.371t/s]  
  
[4/6] Evaluating Configuration: Hybrid_512_5  
  
Evaluating: Hybrid_512_5  
Loading weights: 100%|███████████████████████████████████████| 103/103 [00:00<00:00, 1033.221t/s]  
  
[5/6] Evaluating Configuration: Hybrid_Rerank_512_3  
  
Evaluating: Hybrid_Rerank_512_3  
Loading weights: 100%|███████████████████████████████████████| 103/103 [00:00<00:00, 4785.531t/s]  
Loading weights: 100%|███████████████████████████████████████| 105/105 [00:00<00:00, 1329.941t/s]  
  
[6/6] Evaluating Configuration: Hybrid_Rerank_1024_3  
  
Evaluating: Hybrid_Rerank_1024_3  
Loading weights: 100%|███████████████████████████████████████| 103/103 [00:00<00:00, 5840.621t/s]  
Loading weights: 100%|███████████████████████████████████████| 105/105 [00:00<00:00, 1259.101t/s]
```

```

BENCHMARK RESULTS
=====

Semantic_512_3:
  Context Precision: 0.600
  Answer Relevancy: 0.400
  Faithfulness: 0.200
  Response Time: 0.025s
  Success Rate: 100.0%

Semantic_512_5:
  Context Precision: 1.000
  Answer Relevancy: 0.400
  Faithfulness: 0.200
  Response Time: 0.024s
  Success Rate: 100.0%

Hybrid_512_3:
  Context Precision: 0.740
  Answer Relevancy: 0.800
  Faithfulness: 0.400
  Response Time: 0.058s
  Success Rate: 100.0%

Hybrid_512_5:
  Context Precision: 1.000
  Answer Relevancy: 0.800
  Faithfulness: 0.400
  Response Time: 0.051s
  Success Rate: 100.0%

HybridRerank_512_3:
  Context Precision: 0.600
  Answer Relevancy: 0.900
  Faithfulness: 0.450
  Response Time: 1.399s
  Success Rate: 100.0%

HybridRerank_1024_3:
  Context Precision: 0.600
  Answer Relevancy: 0.900
  Faithfulness: 0.450
  Response Time: 0.899s
  Success Rate: 100.0%

[SUCCESS] Statistical comparison report saved to benchmark_results.json

(calderr-env) PS C:\Users\USER\Desktop\calderr-ai-2026>

```

Feature 2: Streamlit Dashboard

RAG Evaluation Benchmark

Compare, analyze, and optimize multiple RAG configurations with statistical precision.

🚀 Run Evaluation Benchmark

RAG Evaluation Benchmark

Compare, analyze, and optimize multiple RAG configurations with statistical precision.

🚀 Run Evaluation Benchmark

🎉 Evaluation benchmark complete!

🏆 Best Configuration: Semantic_512_5

Context Precision

1.000

Answer Relevancy

0.400

Faithfulness

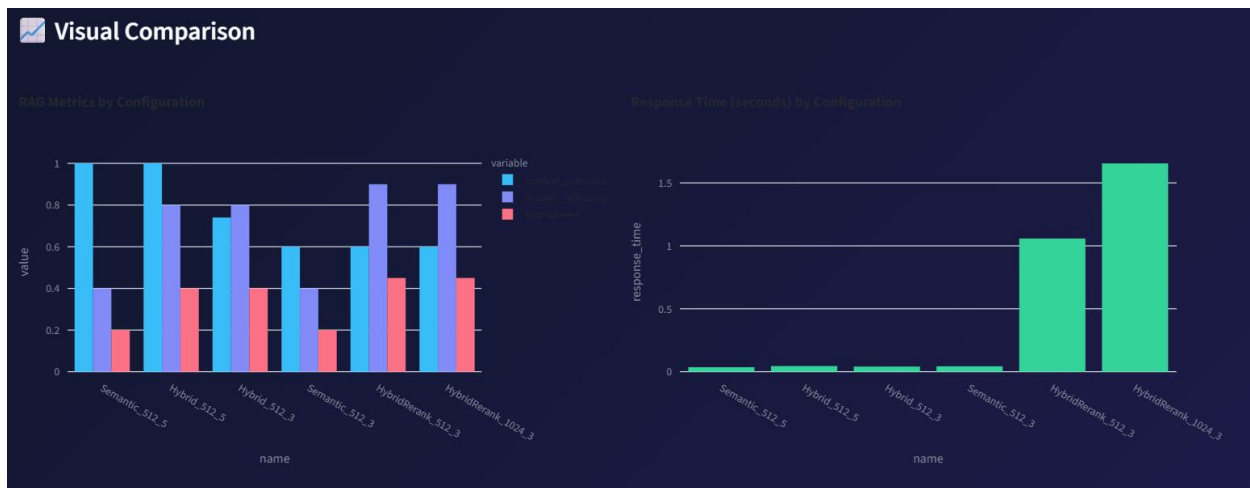
0.200

Response Time

0.036s

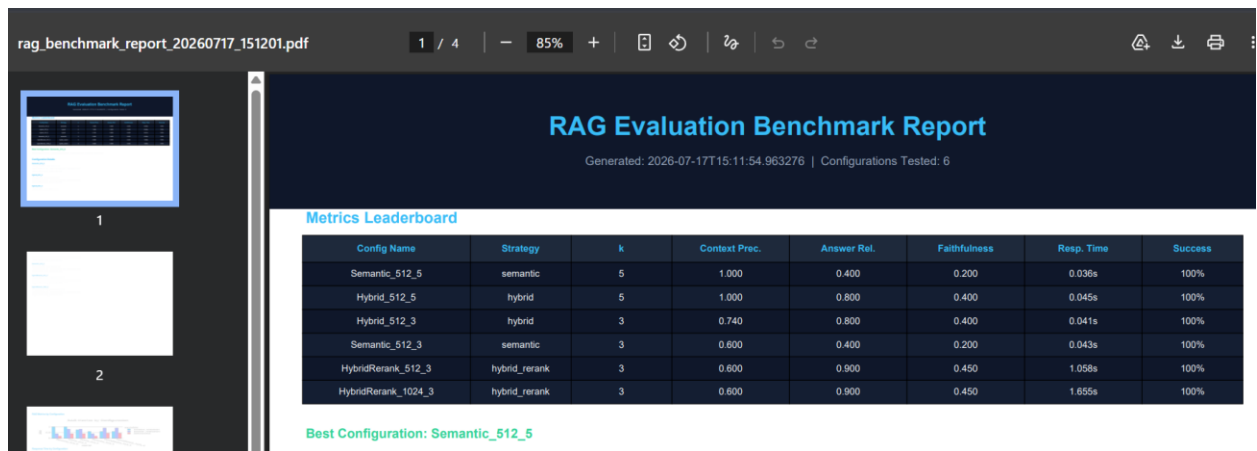
Metrics Leaderboard

	name	chunk_size	chunk_overlap	embedding_model	retrieval_strategy	k	context_precision	answer_relevancy	faithfulness	response_time	success_rate
1	Semantic_512_5	512	50	all-MiniLM-L6-v2	semantic	5	1.000000	0.400000	0.200000	0.036000	1.000000
3	Hybrid_512_5	512	50	all-MiniLM-L6-v2	hybrid	5	1.000000	0.800000	0.400000	0.045000	1.000000
2	Hybrid_512_3	512	50	all-MiniLM-L6-v2	hybrid	3	0.740000	0.800000	0.400000	0.041000	1.000000
0	Semantic_512_3	512	50	all-MiniLM-L6-v2	semantic	3	0.600000	0.400000	0.200000	0.043000	1.000000
4	HybridRerank_512_3	512	50	all-MiniLM-L6-v2	hybrid_rerank	3	0.600000	0.900000	0.450000	1.058000	1.000000
5	HybridRerank_1024_3	1024	100	all-MiniLM-L6-v2	hybrid_rerank	3	0.600000	0.900000	0.450000	1.655000	1.000000



Feature 4: Report Export

a) PDF report



b) JSON report

```
{ } benchmark_results.json > ...
1  {
2    "timestamp": "2026-07-17T15:01:00.923057",
3    "total_configs": 6,
4    "configs": [
5      {
6        "name": "Semantic_512_3",
7        "chunk_size": 512,
8        "chunk_overlap": 50,
9        "embedding_model": "all-MiniLM-L6-v2",
10       "retrieval_strategy": "semantic",
11       "k": 3,
12       "context_precision": 0.6,
13       "answer_relevancy": 0.4,
14       "faithfulness": 0.2,
15       "response_time": 0.025,
16       "success_rate": 1.0
17     },
18     {
19       "name": "Semantic_512_5",
20       "chunk_size": 512,
21       "chunk_overlap": 50,
22       "embedding_model": "all-MiniLM-L6-v2",
23       "retrieval_strategy": "semantic",
24       "k": 5,
25       "context_precision": 1.0,
26       "answer_relevancy": 0.4,
27       "faithfulness": 0.2,
28       "response_time": 0.024,
29       "success_rate": 1.0
30     },
31     {
32       "name": "Hybrid_512_3",
```